FEniCS Conference 2026 Book of Abstracts

FEniCS Conference 2026 Organizing Committee

10th Jun, 2026

Paris, 17-19 June 2026

Total abstracts: **67**

Wednesday - 17 June

Session 1: Software packages with later demonstrations

Time	Contribution	Presenter
13:30 - 13:45	CutFEMx: A Cut Finite Element Library for FEniCSx	Susanne Claus
13:45 - 14:00	Can we predict stress fields without models? Recent adventures of ddfenicsx for Data-driven Identification	Felipe Rocha
14:00 - 14:15	Z3ST: A FEniCSx Framework for Multiscale Thermo-Mechanical Analysis with Automatic Differentiation	Giovanni Zullo
14:15 - 14:30	An Open-Source FEniCSx-Based Platform for Industrial Vibroacoustic Simulation	Antonio Baiano Svizzero
14:30 - 14:45	Automated generation of MITC shell elements in code_aster via FEniCSx symbolic engines	Joaquin Cornejo
14:45 - 15:00	FEniCSx in Metal Additive Manufacturing	Anton Evdokimov
15:00 - 15:15	Numerical investigation of urban heat island within variable porous urban domain	Luis Gerardo Gutierrez Ibarra
15:15 - 15:30	PhiFEM: an immersed boundary finite element method for geometries defined by a level-set	Michel Duprez

Session 2: Kernels, HPC deployment, and the wider FEniCS ecosystem

Time	Contribution	Presenter
16:00 - 16:15	Fast Finite Element Kernels for Modern Computing Architectures	Joseph Dean
16:15 - 16:30	Running after rounding errors: a posteriori error bounds of finite element kernels	Michal Habera
16:30 - 16:45	Distributing FEniCS with the EasyBuild/EESSI ecosystem: how to prepare a software package for distribution in HPC systems	Georgios Kafanas

Time	Contribution	Presenter
16:45 - 17:00	Evolving FEniCS: The Extension Ecosystem at Simula Scientific Computing	Jørgen S. Dokken
17:00 - 17:15	Building a FEniCSx Ecosystem for Biomedical Research	Henrik Finsberg

Posters

Contribution	Presenter
Developing numerical tools to improve the diagnosis of peripheral artery disease	Luke J Barratt
Shape optimization using PhiFEM	Raphaël Bulle
IMPACT - Modelling and optimisation of endless composite filament wound armour plates	Paul T. Kühner
What's new in dolfiny - high-level convenience wrappers for DOLFINx	Andreas Zilian
Adaptive edge element method for a quasilinear problem in electromagnetism with strong convergence for Gauss' law	Sanyang Liu
Flow distortion generation using topology optimization	Langlet Nathan
Effects of Pre-existing Defects on Helium Bubble Nucleation and Growth in Tungsten	Emna Frikha
Finite Element Modeling of Electrochemical Impedance Spectroscopy in All-Solid-State Batteries using FEniCSx	Théo Bermond
GPU kernels in DOLFINx	Chris Richardson
Numerical simulation of asphalt solar collector systems	Lucia Escudero Sartages
Integration of Externally Defined Constitutive Models into FEniCSx Using DOLFINx-External-Operator	Andrey Latyshev

Software demonstrations

Contribution	Presenter
CutFEMx: a cut finite element library for FEniCSx	Susanne Claus
Software demonstration (GPU)	Chris Richardson
Towards natural language-driven computational mechanics: A general LLM-integrated platform built on FEniCS	Guangjin Mou
Numerical investigation of urban heat island within variable porous urban domain	Luis Gerardo Gutierrez Ibarra
Live Demonstration of Z3ST: A FEniCSx Framework for Multiscale Thermo-Mechanical Analysis with Automatic Differentiation	Giovanni Zullo
Live Demonstration of an Open-Source Vibroacoustic GUI Built on FEniCSx	Antonio Baiano Svizzero
FEniCSx in Metal Additive Manufacturing	Anton Evdokimov
From Micro to Macro: Unsupervised FE2 Acceleration using ddfenicsx and micmacsfenicsx	Felipe Rocha
Integration of Externally Defined Constitutive Models into FEniCSx Using DOLFINx-External-Operator	Andrey Latyshev

Thursday - 18 June

Session 3: Multiphysics

Time	Contribution	Presenter
9:00 - 9:15	FELiCS: A Versatile Linearized Flow Solver for Multi-Physics Applications	Marina Matthaiou
9:15 - 9:30	Finite Element Modeling of Electrochemical Impedance Spectroscopy in All-Solid-State Batteries using FEniCSx	Théo Bermond
9:30 - 9:45	Multi-physics Modelling of Nuclear Fusion Reactor Components using FESTIM	James Dark
9:45 - 10:00	FESTIM v2.0: Upgraded framework for multi-species hydrogen transport	Remi Delaporte-Mathurin
10:00 - 10:15	Numerical quality factor statistics of multilayered superconducting radio-frequency cavity	Aaron Gobeyn
10:15 - 10:30	A fully monolithic formulation for multiphase fluid-structure interaction	Marc Hirschvogel

Session 4: Biomedical modelling and applications

Time	Contribution	Presenter
11:00 - 11:15	Nonlinear Inversion Framework - Magnetic Resonance Elastography (MRE)	Henrik Palme
11:15 - 11:30	In-silico drug delivery in the human brain	Cécile Daversin Catty
11:30 - 11:45	Data-integrated simulations of solute transport in the rodent brain using FEniCSx	Andreas Solheim
11:45 - 12:00	Poromechanics to Investigate the Impact of Mechanical Loading on Human Skin Micro-Circulation	Thomas Lavigne
12:00 - 12:15	Contact modeling using the Third Medium Contact (TMC) method: toward the application of a femoral stem insertion inside a bone cavity	Pierre Bichon
12:15 - 12:30	Inferring Human Intracranial CSF Absorption Sites via Inverse Modelling of Protein Transport	Marius Causemann

Session 5: Solid mechanics, contact, and material response

Time	Contribution	Presenter
14:00 - 14:15	Material model discovery with FEMU in FEniCSx	Saeid Ghouli
14:15 - 14:30	Finite-Pressure Anisotropic Hyperelasticity in FEniCSx: A Phenomenological Approach for Shock Physics	Paul Bouteiller
14:30 - 14:45	High-performance solid mechanics simulations in FEniCSx	Musa Choudhury
14:45 - 15:00	Flexibility Method for Contact Mechanics in FEniCSx. Finite Element Construction and Hierarchical Compression of Compliance Operators.	Yahya Boye
15:00 - 15:15	Computational Shakedown Analysis with FEniCSx: Conic Optimization for Large-Scale FEM	Lavakumar Veludandi
15:15 - 15:30	Convergence Study on Polycrystalline Materials	Donald Boyce

Session 6: Solver workflows, inverse modelling, and FEniCS ecosystem

Time	Contribution	Presenter
16:00 - 16:15	Variational Solvers for Irreversible Evolutionary Systems: Orchestration	Andrés A León Baldelli
16:15 - 16:30	Implementation of the TiNiest Tensor de Rham subcomplex on 3D Hybrid Meshes	Johnny Vogels
16:30 - 16:45	An implementation of the distributed two scales method for linear elasticity problems based on FEniCSx and PETSc	Salzman Alexis
16:45 - 17:00	Adjoint-accelerated inverse modelling of acoustic cross-talk in piezoelectric inkjet printheads	Javier Lorente-Macías
17:00 - 17:15	FEniCS Post-processing with ParaView	Louis Gombert
17:15 - 17:30	The Digital Math framework as a robust and general scientific basis for society - advancements in the Adaptive Euler breakthrough for aerodynamics and DigiMat educational program	Johan Jansson

Friday - 19 June

Session 7: Inverse problems, waves, and flow/transport

Time	Contribution	Presenter
9:15 - 9:30	Generalized Stokes problem with pressure Dirichlet boundary conditions for heat transport	Xavier Cartoixà
9:30 - 9:45	Electromagnetic Waveguide Input/Output Boundary Condition using SciFEM and FEniCSx	G. William Slade
9:45 - 10:00	Towards fully programmable inflatable panels	Ofir Mirkin
10:00 - 10:15	A Finite-Element Method for Fluctuating Navier–Stokes Equations	Dimitrios Gourzoulidis
10:15 - 10:30	Flow distortion generation using topology optimization	Langlet Nathan

Session 8: Engineering applications in structures, propulsion, and manufacturing

Time	Contribution	Presenter
11:00 - 11:15	Finite-Element Modeling of Complex Inelasticity Using FEniCSx: Paper and Thermoplastics	Johannes Neumann
11:15 - 11:30	Large scale random vibration analysis using FEniCS	Shubham Saurabh
11:30 - 11:45	femOHL: A FEniCSx-based Framework for Structural Integrity and Resilience Assessment of Overhead Power Transmission Infrastructure	Daria Mesbah
11:45 - 12:00	Multiphysics Eddy-Current Simulation for Efficient Hybrid-Electric Aircraft Propulsion	Arshad Fasiludeen
12:00 - 12:15	L-PBF Processes via FEniCS: A Finite Element Framework for Heat Modelling and Accelerated Predictive Workflows	José Ángel Bejarano Vázquez